

Evaluating predictive count data distributions in retail sales forecasting

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ABSTRACT

Massive increases in computing power and new database architectures allow data to be stored and processed at finer and finer granularities, yielding count data time series with lower and lower counts. These series can no longer be dealt with using the approximative methods that are appropriate for continuous probability distributions. In addition, it is not sufficient to calculate point forecasts alone: we need to forecast entire (discrete) predictive distributions, particularly for supply chain forecasting and inventory control, but also for other planning processes. However, tools that are suitable for evaluating the quality of discrete predictive distributions are not commonly used in sales forecasting. We explore classical point forecast accuracy measures, explain why measures such as MAD, MASE and wMAPE are inherently unsuitable for count data, and use the randomized Probability Integral Transform (PIT) and proper scoring rules to compare the performances of multiple causal and noncausal forecasting models on two datasets of daily retail sales.

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1. Introduction

Count data are integer-valued time series. Count data are especially important for supply chain forecasting, where most products are sold in units. A sufficiently fine granularity along any of the typical dimensions in a demand hierarchy (location, product or time) turns count data into *intermittent demand*, i.e., data that exhibit “many” zeros. For instance, most of the demand series that a typical supermarket faces are intermittent at a store $\times$ SKU $\times$ day level, though of course the series become non-intermittent with aggregation along any dimension.

High-volume count data, e.g., sufficiently highly aggregated data, can and usually will be forecasted using methods that, strictly speaking, are only valid for continuous data, since the relative error between the underlying count data and the continuous approximation will become small

with higher volumes, and many statistical theorems relying on large-sample theory, such as the Central Limit Theorem, can be applied.

However, in recent years the trend has been towards “Big Data”, i.e., the collection and processing of more and more data. Modern database systems allow data of finer and finer granularities to be stored (Januschowski, Kolassa, Lorenz, & Schwarz, 2013). Operational forecasting in retail, e.g., for store replenishment, is already conducted not at highly aggregated levels, but at the most fine-grained level possible; that is, for count data and intermittent demands. As a consequence, count data and intermittent demand forecasting form an active research area, and one that will become even more important in the future; new forecasting algorithms are being developed constantly, together with variations on old ones.

However, so far, the literature on count data and intermittent demand forecasting has focused on point forecasts instead of entire predictive distributions, as has been the norm for many years now in macroeconomic or financial forecasting, for example. We argue that this

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emphasis on point forecasts misses the mark. Instead, we should strive to understand the entire predictive distribution, even for count data, from which we can extract point, interval or quantile forecasts as desired. For example, quantile forecasts are obviously necessary in supply chain forecasting for setting safety amounts, but also for scenario analyses in promotional or other forecasts.

The question now arises as to how different forecasting methods should be compared. For this, one needs forecast accuracy measures, or equivalently, error measures. Count data pose specific challenges for error measures. In particular, minimizing common error measures does not necessarily lead to the “best” forecasting method for count data, especially for intermittent demand series. In addition, while the use of the Probability Integral Transform (PIT) and scoring rules for evaluating continuous predictive distributions is well established, their discrete counterparts have not been used in sales forecasting so far.

This paper is organized as follows: first, we discuss classical point forecast accuracy measures and their problems for count data. Next, we examine a randomization-based modification of the standard PIT, as well as proper scoring rules that are applicable to discrete data. Then, we apply multiple discrete models to two count datasets and evaluate the predictive distributions using the randomized PIT and scoring rules. Finally, we close with possibilities for future research.

2. Point forecast accuracy measures

2.1. Measures based on absolute errors

Assume that the future data generating process that we wish to forecast follows an estimated predictive density $\hat{f}$, and let us assume for now that $\hat{f} = f$ is indeed the true distribution f of future actuals; i.e., our predictive density is correctly specified. It is well known that using the expected value of f as a point forecast will minimize the expected squared error. It is almost as well known (and is an easy derivation to show, cf. Hanley, Joseph, Platt, Chung, & Belisle, 2001; Schwertman, Gilks, & Cameron, 1990) that using the median of f as a point forecast will minimize the expected absolute error.

Let us turn this argument around: the forecast which minimizes (some estimator of) the expected absolute error will estimate not the expected value of f , but its median. In particular, ranking or selecting forecasting methods, or optimizing the parameters of a forecasting method, based on the mean absolute deviation (MAD) will reward point forecasts for yielding not the future expected value, but the future median realization (Morlidge, 2015).

This distinction makes no difference in the case of a symmetric predictive distribution f . However, the predictive distributions that are appropriate for low volume count data (Syntetos, Babai, Lengu, & Altay, 2011) are usually far from symmetric, and this distinction does make a difference in such cases. The fact that optimizing the MAD yields biased forecasts in this case, or even EMAD-optimal forecasts that are constant at zero, has been recognized before (Prestwich, Rossi, Tarim, & Hnich, 2014; Snyder, Ord, & Beaumont, 2012; Teunter & Duncan, 2009; Wallström

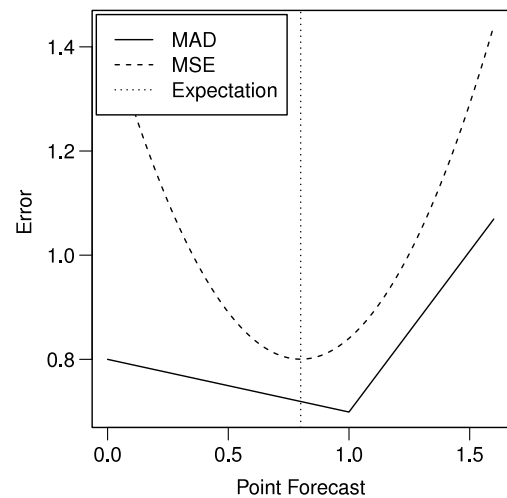


Fig. 1. MADs and MSEs of various point forecasts for $y_1, \dots, y_{10^7} \sim \text{Pois}(0.8)$. The MSE is minimized by the expectation of 0.8, while the MAD is minimized by the median of 1.

& Segerstedt, 2010), but Morlidge (2015) appears to have been the first to explicitly note the connection to the median of f .

As an example, assume that $f = \text{Pois}(\lambda)$ for $\lambda < \log 2 \approx 0.693$. In this case, the median of f is 0, whereas its expectation is λ . The EMAD-optimal point forecast is 0, regardless of whether $\lambda = 0.01$, $\lambda = 0.1$ or $\lambda = 0.5$. Thus, an EMAD-optimal point forecast will be biased downward. Similarly, if $\log 2 < \lambda < \lambda_0$, where $\lambda_0 \approx 1.678$ satisfies $\lambda_0 e^{-\lambda_0} + e^{-\lambda_0} = \frac{1}{2}$, then the EMAD-optimal point forecast will be 1, which is biased upward for $\log 2 < \lambda < 1$ and downward for $1 < \lambda < \lambda_0$.

We note that the preceding example does not in any way presuppose that we constrain the point forecast to be integers. (Indeed, an unbiased point forecast would of course simply be λ itself.) Rather, the argument is that, as described above, the point forecast that minimizes the EMAD is the median of the future distribution—and the median of a Poisson-distributed variable turns out to be integer. Therefore, if our aim is to minimize the EMAD of a Poisson predictive distribution, this cost function automatically draws us towards an integer-valued point forecast, namely the median, which will typically be biased.

To illustrate this point, we calculated the MAD and MSE values of various (not necessarily integer) point forecasts for 10^7 simulated Poisson distributed random variables with parameter $\lambda = 0.8$. As expected, the MSE is minimized by a point forecast of 0.8, i.e., the expectation, while the MAD is minimized by a biased point forecast of 1, the median of the $\text{Pois}(0.8)$ distribution (Fig. 1).

Exactly the same argument applies to all accuracy measures that are multiples of the MAD by a factor that does not depend on the forecast itself. For instance, the weighted mean absolute percentage error (wMAPE) is obtained by dividing the MAD by the mean of the out-of-sample realizations (Kolassa & Schütz, 2007), and the mean absolute scaled error (MASE Franses, 2016; Hyndman, 2006; Hyndman & Koehler, 2006) is obtained

by dividing the MAD by the in-sample MAD of the random walk forecast. Finally, comparing different forecasting methods based on how often each one yields lower absolute errors (the “percentage better” error measure based on MADs) will of course suffer from the same problem.

2.2. Percentage errors

The mean absolute percentage error (MAPE) is undefined if any future realization is zero, so it is singularly unsuitable for count data. However, even if all future realizations are nonzero, there is still the issue that the $E(M)APE$ -optimal forecast will be given by the (-1) -median of the predictive density (Gneiting, 2011), which is generally not the mean, and thus is biased, because of the $(M)APE$'s asymmetric treatment of over- and underforecasts. Kolassa and Martin (2011) provide a very simple pedagogical illustration of a strictly positive discrete symmetric distribution with an explicit biased $E(M)APE$ -optimal forecast. (Leaving count data aside for a moment, we note that the (-1) -median, and thus the $E(M)APE$ -minimal forecast for a lognormal distribution, turns out to be exactly its mode.)

The symmetric MAPE (sMAPE) is an (unsuccessfully, see Goodwin & Lawton, 1999) “symmetrized” version of the MAPE, which is defined if, at all future time points, the point forecasts and actuals are not *both* zero. However, in any period with a zero actual, its contribution is two, regardless of the point forecast (Boylan & Syntetos, 2006), again making it unsuitable for count data.

2.3. Measures based on squared errors

As was noted above, minimizing the squared error naturally leads to an unbiased point forecast. However, Boylan and Syntetos (2006) note that “the mean squared error is not suitable for intermittent-demand items because it is sensitive to the occurrence of very high forecast errors”. The same argument holds for non-intermittent count data.

The geometric root mean square error (GRMSE; proposed by Syntetos & Boylan, 2005) is identical to the geometric mean absolute error (GMAE), since squares and roots cancel out in geometric means (Hyndman, 2006). Thus, the bias of an $EGRMSE$ -optimal point forecast for an asymmetric f will depend on the specific shape of f , but will be lower than for an $EMAD$ -optimal forecast.

However, if one of the absolute errors is zero, the GRMSE will be zero, regardless of how large the other absolute errors are (Boylan & Syntetos, 2006). This will happen if any actual is forecasted correctly, which can easily happen in intermittent demand. Boylan and Syntetos (2006) focus on the naïve random walk forecast, which will indeed exhibit this behavior if two or more zero actuals follow each other, and claim that the naïve forecast is inappropriate for intermittent demand – which Hyndman (2006) disputes – and can therefore be disregarded. However, we submit that this argument begs the question. After all, we are using error measures *precisely* in order to distinguish more appropriate from less appropriate forecasting methods. In addition, the GRMSE also exhibits this shortcoming for any other forecast that happens to hit a single actual exactly, e.g., a constant zero or a constant one forecast.

2.4. Relative errors

Relative errors measure the error of a given forecast *relative* to a benchmark forecast, e.g., the naïve random walk forecast. Prominent variations are the median relative absolute error (MdRAE) and the geometric mean relative absolute error (GMRAE). In the specific context of forecasting count data, these suffer from two main weaknesses:

- Relative errors commonly compare absolute errors. As such, they are subject to the same criticisms as MAD-based errors, as detailed above.
- On a period-by-period basis, simple benchmarks such as the naïve random walk may forecast without error, and thus, this period's relative error would be undefined because of a division by zero (Hyndman, 2006).

The second weakness could be addressed to some extent by taking ratios of summary errors over multiple periods rather than calculating summaries over period-by-period relative errors; e.g., dividing the sum of the absolute errors of the forecast in question by the sum of the absolute errors of the random walk. This would only be undefined if the random walk had no errors over the entire forecast horizon, which is an uncommon occurrence unless we have a very short forecasting horizon.

2.5. Rate-based errors

Kourentzes (2014) recently proposed two new error measures for intermittent demand, the MSR and the MAR, which aim to assess whether an intermittent demand point forecast captures the average demand correctly over an increasing period of time. This is an interesting proposal, but one property of these measures is that they implicitly weight the near-term future more heavily than the longer-term future. One could argue that this is exactly what we want to do in forecasting, but even then a case could be made that such a weighting should be explicit, e.g., by using an appropriate weighting scheme when averaging over future time periods.

2.6. Scaled errors

Petropoulos and Kourentzes (2015) propose a scaled version of the MSE, the sMSE, which is the mean over squared errors that have been scaled by the squared average actuals over the forecast horizon. The sMSE is well-defined unless all actuals are zero, is minimized by the expectation of f and, due to the scaling, can be compared between different time series. In addition, again because of the scaling, it is not quite as sensitive to high forecast errors as the MSE (Boylan & Syntetos, 2006); specifically, it is more robust to dramatic *underforecasts*, although it is still sensitive to large *overforecasts*. Overall, it appears to be the most promising candidate for the evaluation of the point forecast accuracy for count data.

2.7. Functionals and loss functions

An alternative way of looking at forecasts concentrates on point forecasts that are functionals of the predictive

distribution (Gneiting & Katzfuss, 2014). One could argue that a retailer aims at a certain service level, say 95%, and that therefore he is only interested in the corresponding quantile of the predictive distribution. This would then be elicited (Lambert, Pennock, & Shoham, 2008) through the use of appropriate loss functions or scoring rules. This approach is closely related to the idea of considering forecasts as part of a stock control system (e.g., Syntetos, Babai, & Gardner, 2015). In this perspective, quantile forecasts are used as inputs to standard stock control strategies, and the quality of the forecasts is assessed by valuing the total stock position over time and weighting it against out-of-stocks.

On the one hand, this argument has merit. Considering predictive quantiles directly is certainly a step forward from setting stock targets based on mean forecasts plus a multiple of the forecasted standard deviation, implicitly invoking an inappropriate normal distribution, as retailers frequently do. On the other hand, simple quantiles over single periods are not enough.

In store replenishment, retailers need to consider logistical constraints (pack sizes, best by dates, delivery schedules, minimum or maximum truck loads), balancing a lower service level for one product against a higher service level for another. They need to optimize over complex cost functions (time-varying purchasing and sales prices, rebate brackets that kick in once a certain total order amount has been reached), which in turn require flexibility with regard to lead times: we may want to pull orders forward in time, a so-called “forward buy”. In addition, frequently retailers are also interested in quantile forecasts at different aggregation levels: store $\times$ SKU level forecasts are important for store-level replenishment, but *aggregate* forecasts across stores are necessary when ordering stock from the manufacturer centrally, and forecasts for aggregates across both SKUs and stores are important for planning promotions, which typically involve multiple SKUs and multiple stores. All of these considerations argue strongly against aiming at single functionals of the predictive distribution and for attempting to obtain a correct overall predictive distribution.

2.8. Predictive distributions and their assessment

Willemain (2006) recommends forecasting not only average demands, but the entire distribution of future demands, and suggests that the accuracy of such a predictive distribution be assessed using a χ^2 statistic. While focusing on predictive distributions instead of point forecasts makes a lot of sense, particularly for count data and intermittent demands (see above), the χ^2 error measure, or any other measure that focuses solely on period-by-period errors, has one fundamental shortcoming: it presupposes that both the future actuals and the predictive distributions are independent and identical over time. However, even count data can exhibit nonstationarity, e.g., spare parts demands may be dominated by life cycles or seasonality, and retail sales may be driven by promotions and days of the week.

Snyder et al. (2012) also emphasize that more attention should be paid to forecasting the full discrete distribution, instead of only a point forecast. They propose two scores

for evaluating discrete predictive distributions, namely the prediction likelihood score (PLS), which is the exponential of the logarithmic score described below, and the discrete ranked probability score (DRPS), which measures the L_2 distance between the proposed and actual future distributions.

Boylan and Syntetos (2006) note that measuring the accuracy of a whole predictive distribution may yield misleading information: a predictive distribution may appear very accurate because it gets the lower part of the future distribution right, while breaking down for higher quantiles, which is often the part of the future distribution that we are most interested in, in order to define safety amounts. Consequently, Boylan and Syntetos (2006) propose the evaluation of quantile forecasts of a small set of predefined high critical values, e.g., 80%, 90% and 95%, with exceedances being analyzed using a χ^2 test. This approach would also work with non-i.i.d. actual and predictive distributions. However, in addition to the problems associated with focusing on single functionals, as described in the previous section, if we consider a classical null hypothesis significant testing framework, the p -values obtained using *multiple* critical values would need to be subjected to corrections for dependent multiple comparisons. Also, this approach would not merely downweight, but completely disregard the lower tail of the predictive distribution, thus losing a lot of power.

We note that the point raised by Boylan and Syntetos (2006) highlights one potential criticism of the PLS and DRPS proposed by Snyder et al. (2012): they can indicate whether a proposed forecasting algorithm is better or worse than a competing algorithm for forecasting the future density, but offer little guidance as to where exactly the predictive distributions differ.

3. Assessing discrete predictive distributions

We fully concur with the recommendations of Snyder et al. (2012) and Willemain (2006) that the focus should be on the entire predictive distribution, instead of only the point forecast. On the one hand, the problems with the point forecast accuracy measures described above are serious. On the other hand, especially for supply chain forecasting, a single point forecast for a particular functional is not sufficient. When setting safety stocks, we need quantiles, which argues for at least interval forecasts. If we also need to set safety stocks for complex multi-period lead times, we will need to handle quantiles of sums of random variables, which requires us to understand the full predictive distribution. However, as was mentioned at the end of the previous section, there is not yet a good understanding of ways of evaluating the quality of a given discrete predictive distribution.

3.1. The randomized probability integral transform (rPIT)

One standard method of evaluating *continuous* density forecasts involves the *probability integral transform* (PIT, introduced originally by Rosenblatt, 1952, and independently by J. H. Curtiss and I. R. Savage). Assume that at a (future) time t , we forecast a density $\hat{f}_t$ with cumulative

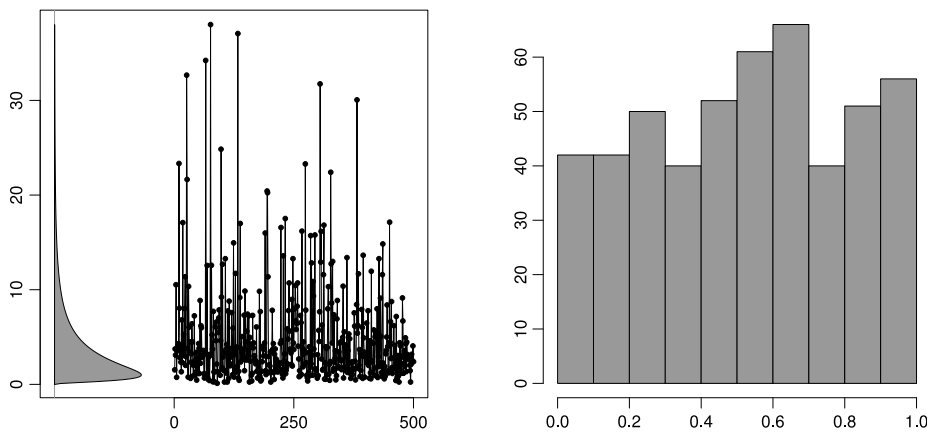


Fig. 2. Left: a time series of 500 simulated lognormal actuals (mean zero and standard deviation one on the log scale), together with the density. Right: a histogram of p values from applying the probability integral transform to a correctly-specified predictive density for these actuals.

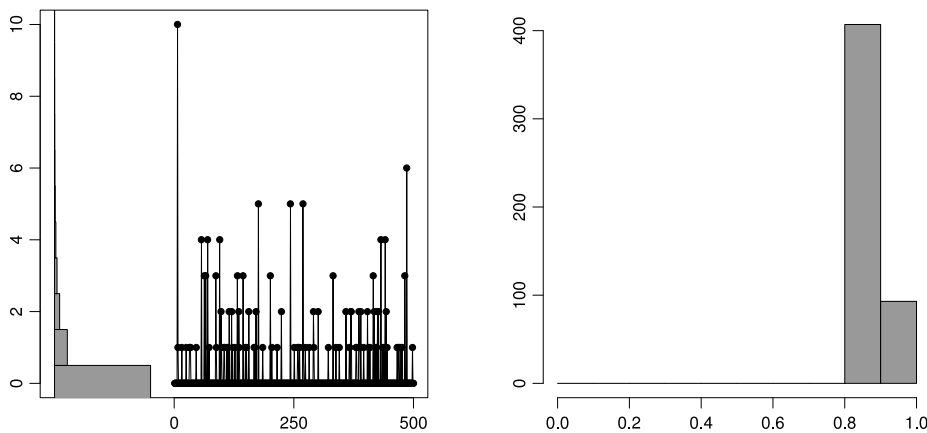


Fig. 3. Left: a time series of 500 simulated negative binomial actuals (mean 0.4, size parameter 0.2), together with the probability mass. Right: a histogram of p values from applying the standard probability integral transform to a correctly-specified predictive density for these actuals.

forecast distribution $\hat{F}_t$, whereas the true cumulative distribution is F_t . Assume further that we observe y_t . Then, the PIT is

$$p_t = \hat{F}_t(y_t) = \int_{-\infty}^{y_t} \hat{f}_t(y) dy. \quad (1)$$

If the forecast density is correct, $\hat{F}_t = F_t$, the p_t are independently identically uniformly distributed on $[0, 1]$, which can be assessed using either graphs or formal tests (for an overview, see [Tay & Wallis, 2000](#)). [Fig. 2](#) illustrates this approach.

Unfortunately, the PIT cannot be transferred directly to discrete distributions. Indeed, assume that we predict a time-invariant discrete predictive distribution that happens to be correct, $\hat{F} = F$. In this case, the PIT $p_t = \hat{F}(y_t)$ will be discrete, since $\hat{F} = F$ is a step function as the cumulative distribution function for a discrete random variable. More specifically, the PIT will be definitely non-uniform, even though the density was forecast correctly. [Fig. 3](#) illustrates the problem with this approach.

To address this problem, a *randomized PIT* (rPIT) has been proposed ([Brockwell, 2007](#); [Frühwirth-Schnatter, 1996](#); [Liesenfeld, Nolte, & Pohlmeier, 2006](#); [Smith, 1985](#)).

If the predictive density is discrete, then instead of the continuous PIT in [Eq. \(1\)](#), we draw each p_t at random from the following uniform distribution:

$$\tilde{p}_t \sim U[\hat{F}_t(y_t - 1), \hat{F}_t(y_t)]. \quad (2)$$

If $\hat{F}_t = F_t$ is specified correctly, the $\tilde{p}_t$ are again i.i.d. on $U[0, 1]$, as their distribution is a mixture of uniform distributions on subintervals of $[0, 1]$, with weights corresponding to the lengths of these subintervals (see [Lemma 2](#) of [Brockwell, 2007](#), or the Appendix of [Liesenfeld et al., 2006](#)), so that again we can use the machinery developed for the continuous case. [Fig. 4](#) illustrates the application of the rPIT to discrete data.

Incidentally, [Czado, Gneiting, and Held \(2009\)](#) propose plotting, not a histogram of randomly sampled $(\tilde{p}_t)$, but their probability mass function, which can be derived from $\hat{F}$ without needing recourse to randomization. This may be of interest in cases where low numbers of future realizations render the rPIT too variable.

Now, how should we assess whether our $\tilde{p}_t$ are i.i.d. on $U[0, 1]$? In the context of count data forecasting for supply chain applications, multiple aspects need to be kept in mind. On the one hand, high quantiles

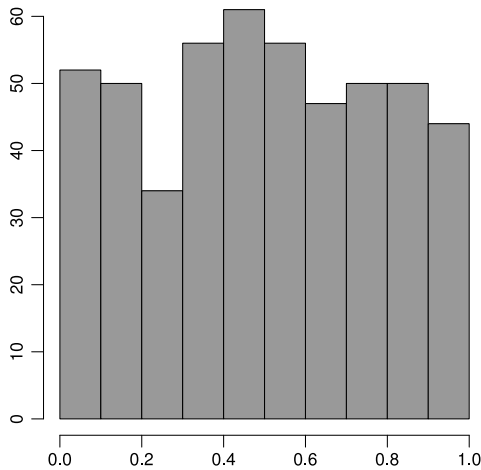


Fig. 4. A histogram of p values from applying the randomized probability integral transform to the negative binomial actuals shown in Fig. 3.

of the predictive density are important for inventory planning, while measures of the goodness-of-fit may be dominated by good fits in lower quantiles, masking an $\hat{F}_t$'s unacceptable fit for higher quantiles (Boylan & Syntetos, 2006). Then again, assessing only the exceedance of a 95% quantile, for example, by assessing interval rather than density forecasts, loses a lot of information. In addition, when dealing with supply chain time series, we are frequently faced with large numbers of time series that require forecasting: thousands of SKUs, possibly in multiple locations, or even hundreds or thousands of locations, as in the case of retail sales. In such a situation, examining each individual series' q-q-plot, or similar approaches (Tay & Wallis, 2000), is no longer feasible. Instead, we need a way of assessing multiple forecast algorithms quickly on large data sets, including detecting unacceptable performances in high quantiles without being misled by good performances in lower quantiles.

We propose the use of a *data driven smooth test* to assess whether $\tilde{p}_t \sim U[0, 1]$. Specifically, we propose to apply the data driven version of Neyman's (1937) smooth test of goodness-of-fit, as introduced by Inglot and Ledwina (2006) and Ledwina (1994). This test, when applied to one time series' vector of $\tilde{p}_t$ s, yields a test statistic WT, for which we can in turn calculate a P value numerically. In addition, we note that this test in fact combines k different test statistics (with k being chosen based on the data, using the BIC), which are sensitive to departures from normality in different regions of $[0, 1]$; thus, we could investigate the k th substatistic in order to assess the performance for specific quantiles, possibly after constraining k to be at least a given minimal value, to achieve sensitivity in high quantiles.

The logical question, then, is how both the $\tilde{p}_t$ s and the results of this smooth test can be summarized for many time series, as supply chains usually involve thousands of SKUs that need to be forecasted at multiple (again, possibly thousands of) locations. We propose two complementary procedures: first, one can simply stack the vector of $\tilde{p}_t$ s for all time series and investigate their distribution graphically, e.g., using histograms. Similarly, we propose

investigating the set of WT statistics that come from the different time series, again summarizing them graphically or calculating appropriate summary or inferential statistics for them. Such analyses can be performed for comparing different forecasting algorithms, different locations, or different product groups.

The literature on evaluating continuous densities has one additional focus: whether or not forecast errors exhibit serial dependence. Of course, this question also arises for count data; however, we suspect that tests for serial correlation (where the errors may well stem from misspecified discrete predictive distributions) will have rather low power for count data. We therefore elected not to pursue this question. A few ideas for future research can be found below.

3.2. Proper scoring rules

We note that a uniformly distributed PIT is a necessary but not sufficient condition for a density forecast to be ideal. Indeed, Hamill (2001) and Gneiting, Balabdaoui, and Raftery (2007) give examples of forecasters who have uniform PITs, but all of whose conditional mean forecasts are biased. This illustrates that a uniform PIT serves to indicate the *calibration* of a density forecast, i.e., its consistency with the actual realizations, but not its *sharpness*, i.e., the concentration of the predictive distributions (Gneiting et al., 2007).

(Proper) scoring rules have recently been introduced to address this shortcoming of the PIT, i.e., as tools for assessing both the calibration and sharpness of predictive distributions (Gneiting et al., 2007; Gneiting & Katzfuss, 2014). A *scoring rule* is a function s that maps a predictive distribution $\hat{f}$ and a single realization y to a penalty value $s(\hat{f}, y)$. In practice, one usually reports averages of the scores over suitable pairs $(\hat{f}, y)$, e.g., over forecasts and actuals over multiple time points t ,

$$S := \frac{1}{T} \sum_{t=1}^T s(\hat{f}_t, y_t).$$

A scoring rule is said to be *proper* if its expected value is minimal if $\hat{f}$ is the true future distribution of y ,

$$E_{y \sim f}(s(\hat{f}, y)) \leq E_{y \sim f}(s(\hat{f}, y)),$$

and *strictly proper* if its expectation is minimized *only* by the true future distribution,

$$E_{y \sim f}(s(\hat{f}, y)) < E_{y \sim f}(s(\hat{f}, y)) \quad \text{if } \hat{f} \neq f.$$

Czado et al. (2009) discuss a number of proper scoring rules in the context of assessing discrete predictive distributions. Assume that our discrete predictive probability mass in forecasting a given observation is given by $(\hat{f}_k)$ and the cumulative predictive probability mass by $(\hat{F}_k)$ for possible realizations $k = 0, \dots, \infty$, and that y is our actual future observation.

- The *logarithmic score* is defined as

$$\text{logs}(\hat{f}_k, y) := -\log \hat{f}_y.$$

- The *predictive deviance* is defined as

$$\text{dev}(\hat{f}_k, y) := -2 \log \hat{f}_y + 2 \log g_y$$

for a suitable standardizing term (g_k).

Both logs and dev are only defined if $\hat{f}_k > 0$ for all k that are actually observed. This is not necessarily given, particularly if $\hat{f}$ is derived not from a parametric model, but from empirical distributions, either by a direct tabulation of historical data, or through simulation-based approaches. We note in passing that the PLS proposed by Snyder et al. (2012), which is the exponential of logs, is analogously zero if a single future realization has a zero probability under the predictive distribution. Since we will make heavy use of simulation-based and empirical approaches in our analyses below, we will not pursue logs, dev or the PLS further.

Let $\|\hat{f}\|^2 := \sum_{k=0}^{\infty} \hat{f}_k^2$, which can often be calculated analytically, e.g., if $\hat{f}$ describes a Poisson (Laurent, 2015b) or negative binomial (Laurent, 2015a) probability mass.

- The *Brier score* (Brier, 1950) or *quadratic score* is defined as

$$\text{qs}(\hat{f}_k, y) := -2\hat{f}_y + \|\hat{f}\|^2.$$

- The *spherical score* is defined as

$$\text{sphs}(\hat{f}_k, y) := -\frac{\hat{f}_y}{\|\hat{f}\|}.$$

- The *ranked probability score* is defined as

$$\text{rps}(\hat{f}_k, y) := \sum_{k=0}^{\infty} (\hat{F}_k - \mathbf{1}_{y \leq k})^2.$$

This was used by Snyder et al. (2012) to evaluate intermittent demand predictive distributions based on a simulation approach.

- The *Dawid-Sebastiani score* (Wei & Held, 2014) is defined as:

$$\text{dss}(\hat{f}_k, y) := \left(\frac{y - \hat{\mu}}{\hat{\sigma}} \right)^2 + \log \hat{\sigma}^2,$$

where $\hat{\mu}$ and $\hat{\sigma}$ are the mean and the standard deviation of $(\hat{f}_k)$.

As was discussed above, all of these can be summarized by averaging, and the average scores of different forecasting methods (yielding different predictive masses $\hat{f}$) can be compared using Diebold–Mariano-type tests (Gneiting & Katzfuss, 2014). However, as was discussed above, these scoring rules are susceptible to the criticism raised by Boylan and Syntetos (2006): they indicate whether a proposed forecasting algorithm is better or worse than a competing algorithm for forecasting the future density, but offer little guidance as to where exactly the predictive distributions differ. In contrast, the rPIT approach described above can be used to obtain this information by assessing each component of Neyman's test applied to $(\tilde{p}_t)$ separately.

4. Empirical analysis

4.1. Data

We tested our approach of assessing discrete forecast distributions on two sets of count data.

Dataset A contains daily sales of 1000 stock keeping units (SKUs) in one drugstore. This retailer uses an every day low price strategy, i.e., he does not use price promotions, and therefore sales prices are irrelevant as causal factors. No seasonality beyond day-of-the-week effects is evident in the aggregate sales plots. Category information is not available.

Dataset B contains daily sales of 1394 grocery SKUs in one supermarket, together with prices. Category information is available, and the 1394 SKUs stem from selected categories that did not exhibit seasonality in aggregate plots. We transformed the prices into percentage price reductions with respect to prices over the previous two weeks, disregarding price increases, reasoning that, on the one hand, long-term price reductions lose their appeal after a few weeks, and on the other, price increases are rarely advertised to consumers.

Fig. 5 shows the intermittency/lumpiness classifications of the SKUs in the two datasets, according to Syntetos, Boylan, and Croston (2005) and Boylan, Syntetos, and Karakostas (2008). While there are differences between the datasets, both exhibit a high degree of intermittency, which motivates our inclusion of Croston-type models below, and most of the time series lie in the three quadrants in which the Syntetos–Boylan approximation works well, according to Boylan et al. (2008) and Syntetos et al. (2005).

4.2. Software

All of the analyses were run in R (R Core Team, 2015). The `gs1` package (version 1.9–10, see Hankin, 2006) was used in the analytical evaluation of the Brier and spherical scores for the Poisson and negative binomial models. The other R packages used are indicated below.

4.3. Models

We applied the following fourteen models to our data (abbreviated as shown below), motivated by Syntetos et al. (2011).

Empirical (E): Discrete predictive densities are given by the empirical probability distribution of all historical sales.

Empirical + Weekdays (E+W): Discrete predictive densities are given by the empirical probability distribution of historical sales on the weekday that we wish to forecast for.

For the next two models, we fit Croston's (1972) model for intermittent time series, using the `crost` function of Kourntzes and Petropoulos (2014) with default parameters, apart from `type="sba"`, which yields point forecasts.

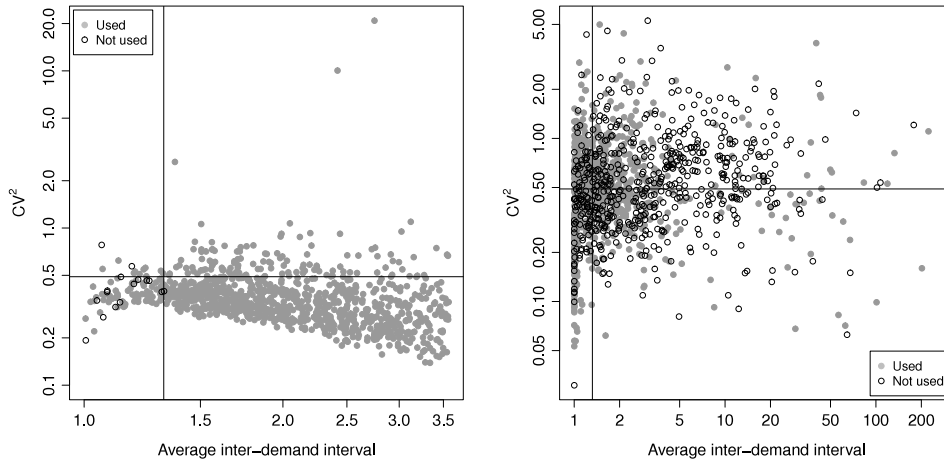


Fig. 5. Intermittency/lumpiness classifications of SKUs in datasets A (left) and B (right), with data points to indicate whether or not the time series was used in the analysis, as per the exclusion criteria noted in the text. The horizontal lines are at $CV^2 = 0.49$, the vertical ones at average inter-demand intervals of 1.32, as per [Boylan et al. \(2008\)](#) and [Syntetos et al. \(2005\)](#). Note the use of log scales.

Croston + Poisson (C+P): Discrete predictive distributions are defined as the Poisson distribution, using the point forecast as the Poisson parameter.

Croston + NegBin (C+NB): We estimate the size parameter of a negative binomial distribution as

$$\hat{s} := \frac{1}{T} \sum_{t=1}^T \frac{\hat{y}_t^2}{\text{var}(y - \hat{y}) - \hat{y}_t}, \quad (3)$$

where $\hat{y}_t$ are the in-sample fits of the Croston model. We define our discrete predictive distribution as a negative binomial distribution, with a mean given by the Croston point forecast and size $\hat{s}$. If Eq. (3) is not defined or yields a negative result, we fall back on the C+P model defined above.

Poisson (P): Discrete predictive densities are defined as the Poisson distribution, using the overall historical average as the parameter.

Bootstrapped Poisson (BP): We bootstrap our data. For each bootstrap replicate, we determine the average sales and then simulate from a Poisson distribution, using this average as a parameter. The discrete predictive distribution is then defined as the empirical distribution of these bootstrapped simulations.

For regression models, we included a Boolean Christmas predictor, covering the last few days before Christmas, as well as a linearly increasing trend predictor, since some categories in dataset B exhibited trends. In the interest of simplicity, the trends are not damped for forecasting. For the next two models, we fit Poisson regression models with a log link function, using weekdays, Christmas, trend and price reduction (if applicable) as independent variables, using the `glm` function in the `stats` package.

Poisson Regression (PR): The discrete predictive distribution is given by a Poisson distribution, using the forecasted response as a parameter.

Bootstrapped Poisson Regression (BPR): We bootstrap our data. For each bootstrap replicate, we fit a Poisson regression and simulate from the fit. The discrete predictive distribution is defined as the empirical distribution of these bootstrap simulations.

NegBin (NB): We fit a negative binomial distribution to our data by matching moments (since R's `fitdistr` function from the MASS package (version 7.3–42, see [Venables & Ripley, 2002](#)), hung on extreme but valid intermittent series), yielding discrete predictive distributions.

Bootstrapped NegBin (BNB): We bootstrap our data. For each bootstrap replicate, we fit a negative binomial model by matching moments, and simulate from the fit. The discrete predictive distribution is defined as the empirical distribution of these bootstrap simulations.

For the next two models, we fit negative binomial regression models, using weekdays, Christmas, trend and price reduction (if applicable) as independent variables, using the `glm.nb` function in the MASS package (version 7.3–42, see [Venables & Ripley, 2002](#)), which estimates a log link function for the mean parameter and a single overdispersion parameter that does not depend on any independent variables.

NegBin Regression (NBR): The discrete predictive distribution is given by a negative binomial distribution, using the forecasted response as the mean parameter, as well as the estimated overdispersion parameter.

Bootstrapped NegBin Regression (BNBR): We bootstrap our data. For each bootstrap replicate, we fit a negative binomial regression as in the NBR model above, and simulate from the fit. The discrete predictive distribution is defined as the empirical distribution of these bootstrap simulations.

The bootstrapped variants were included in order to include the effects of the propagation of parameter uncertainty into the forecasts, since no closed form

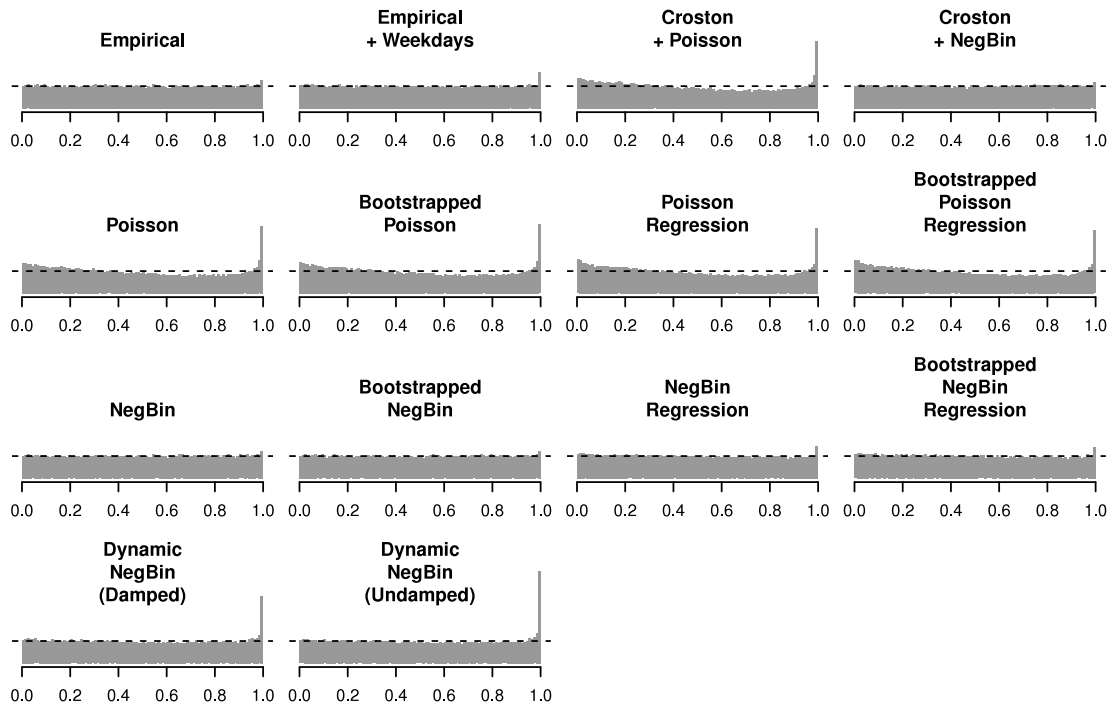


Fig. 6. $\tilde{p}_t$ values over all SKUs from the randomized probability integral transform for dataset A. The vertical axes (not shown) are to a common scale. The horizontal dashed lines indicate uniform distributions.

for predictive distributions that is analogous to using t distributions in ordinary least squares appears to exist.

Finally, we examined two dynamic models that have performed well in the past (Snyder et al., 2012):

Dynamic NegBin Damped (DNBD): We calculate the overall mean historical sales $\mu = \mu_1$, and calculate a smoothed estimate of historical average demands via $\mu_t = (1 - \phi - \alpha)\mu + \phi\mu_{t-1} + \alpha y_{t-1}$, regularized towards μ , where the smoothing parameters $0 < \alpha, \phi < 1$, with $\alpha + \phi < 1$, together with the NegBin overdispersion parameter, are estimated via maximum likelihood. The discrete predictive distribution is given by a negative binomial distribution, using the last smoothed mean μ_T as the mean parameter, as well as the estimated overdispersion parameter.

Dynamic NegBin Undamped (DNBU): We calculate a smoothed estimate of historical average demands via $\mu_t = (1 - \alpha)\mu_{t-1} + \alpha y_{t-1}$ (initializing μ_1 with the overall historical mean), where the smoothing parameter $0 < \alpha < 1$ and the NegBin overdispersion parameter are estimated via maximum likelihood. The discrete predictive distribution is given by a negative binomial distribution, using the last smoothed mean μ_T as the mean parameter, as well as the estimated overdispersion parameter.

4.4. Procedure

For each SKU, we kept the last 100 days as a holdout sample and fitted the models to all previous sales. If

fewer than 300 historical data points were available for an SKU, it was skipped. Each SKU $\times$ model combination was modeled and forecasted a single time (no rolling procedure), since our models took a long time to run, especially the computationally intensive BNBR, DNBD and DNBU models. Each bootstrapped model was bootstrapped 1000 times (Canty & Ripley, 2015).

4.5. Evaluation

On the one hand, we stacked the $\tilde{p}_t$ values resulting from the rPIT and investigated them graphically through histograms.

On the other hand, we subjected each time series' $\tilde{p}_t$ s to a data-driven smooth test for uniformity, using the `ddst.uniform.test` function of Biecek and Ledwina (2012). In light of their strongly skewed distribution, the resulting WT test statistics, one per time series, were first logged, then scatterplotted, beanplotted (Kampstra, 2008), and subjected to pairwise t tests.

Finally, we calculated Brier and spherical scores by averaging the relevant scores over the forecasts of each time series. The scores were again scatter- and beanplotted and subjected to pairwise t tests.

4.6. Results

Skipping SKUs with too little data, as described above, left us with 984 SKUs in dataset A and 820 in dataset B.

Figs. 6 and 7 provide histograms of $\tilde{p}_t$ values over all SKUs and scatter-/beanplots of WT statistics for each SKU, respectively, for dataset A. Figs. 8 and 9 provide analogous information for dataset B.

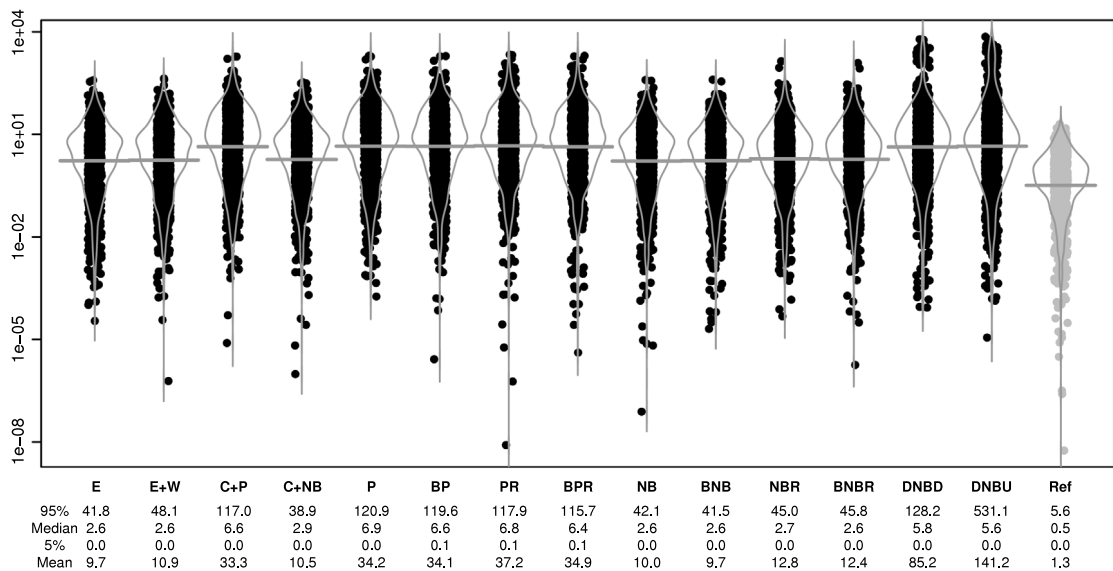


Fig. 7. WT values for each SKU from the randomized probability integral transform for dataset A, as well as for a reference set of length-matched $U[0, 1]$ vectors. Note the logarithmic scale.

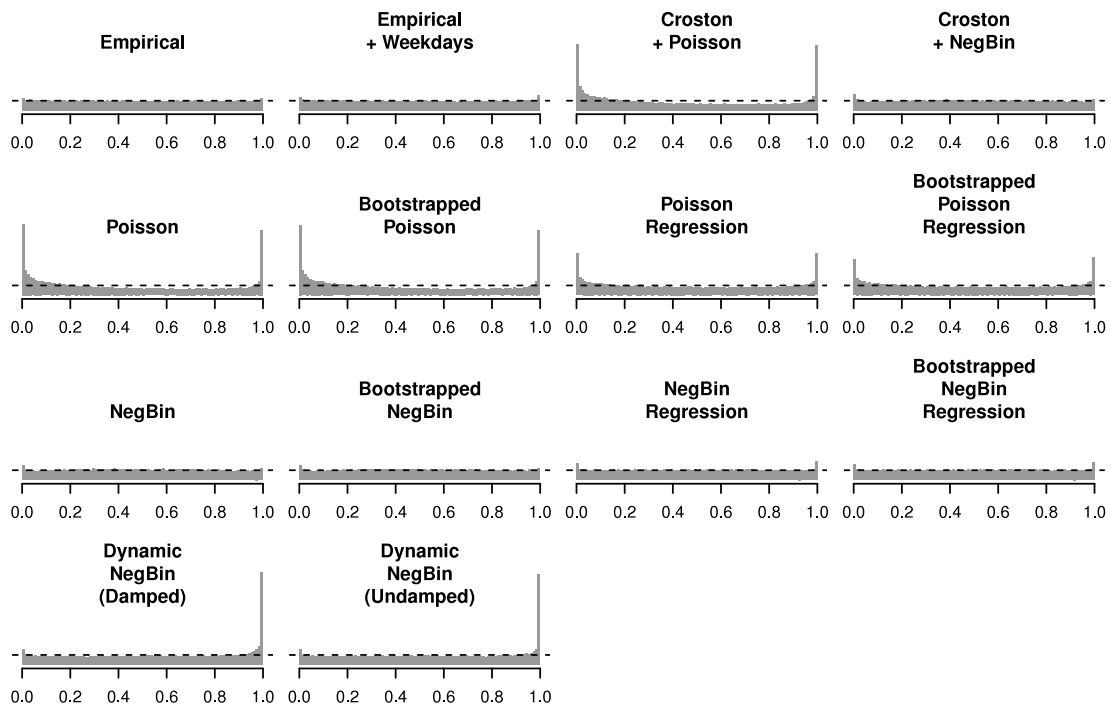


Fig. 8. $\tilde{p}_i$ values over all SKUs from the randomized probability integral transform for dataset B. The vertical axes (not shown) are to a common scale. The horizontal dashed lines indicate uniform distributions.

Table 1 shows the p -values of pairwise paired t -tests between logged WT statistics, corrected for multiple testing (within each dataset) following the sequential procedure of Holm (1979), using the `pairwise.t.test` function in the `stats` package.

Figs. 10 and 11 provide Brier scores for datasets A and B. Table 2 shows p -values of pairwise paired t -tests between Brier scores.

We note that some of the models yield mean logged WT statistics or Brier scores that are not statistically significantly different from each other. Table 3 shows the models ordered by mean WT statistics, with consecutive models that do not yield significantly different mean logged WT statistics being clustered together, and the same information for Brier scores.

Plots, pairwise t -tests and clusterings for spherical scores look qualitatively similar, and are given in an

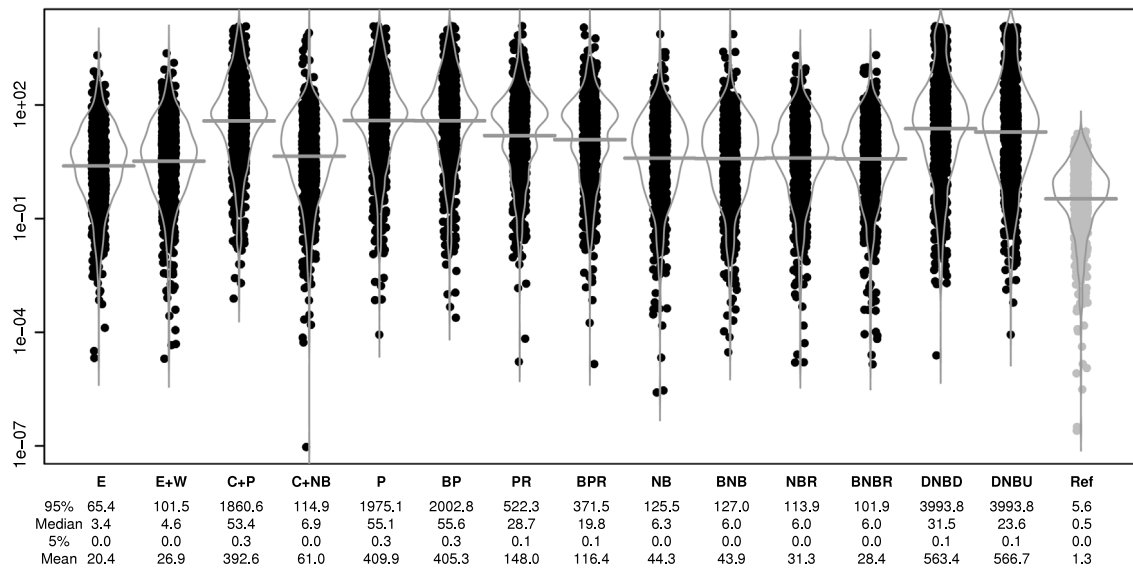


Fig. 9. WT values for each SKU from the randomized probability integral transform for dataset B, as well as for a reference set of length-matched $U[0, 1]$ vectors. Note the logarithmic scale.

Table 1

p-values of pairwise paired *t*-tests between logged WT statistics for the two datasets, corrected for multiple testing.

	E	E+W	C+P	C+NB	P	BP	PR	BPR	NB	BNB	NBR	BNBR	DNBU
E+W	A 1 B .												
C+P	A . B .	*											
C+NB	A 1 B .	1	*										
P	A . B .	*	1	*									
BP	A . B .	*	1	*	1								
PR	A . B .	*	1	*	1	1							
BPR	A . B .	*	1	*	1	1	1						
NB	A 1 B .	1	*	0.36 0.67	*	*	*	*					
BNB	A 1 B .	1	*	1	*	*	*	*	1				
NBR	A 1 B 0.0002	1	*	1	*	*	*	*	1	1			
BNBR	A 1 B 0.0006	1	*	1	*	*	*	*	1	1	1		
DNBD	A . B .	*	1	0.0005	1	0.0002	0.0005	0.004	*	*	*	*	*
DNBU	A . B .	*	1	*	1	1	1	0.83	0.001	*	*	*	1 0.12

* Denotes $p < 0.0001$.

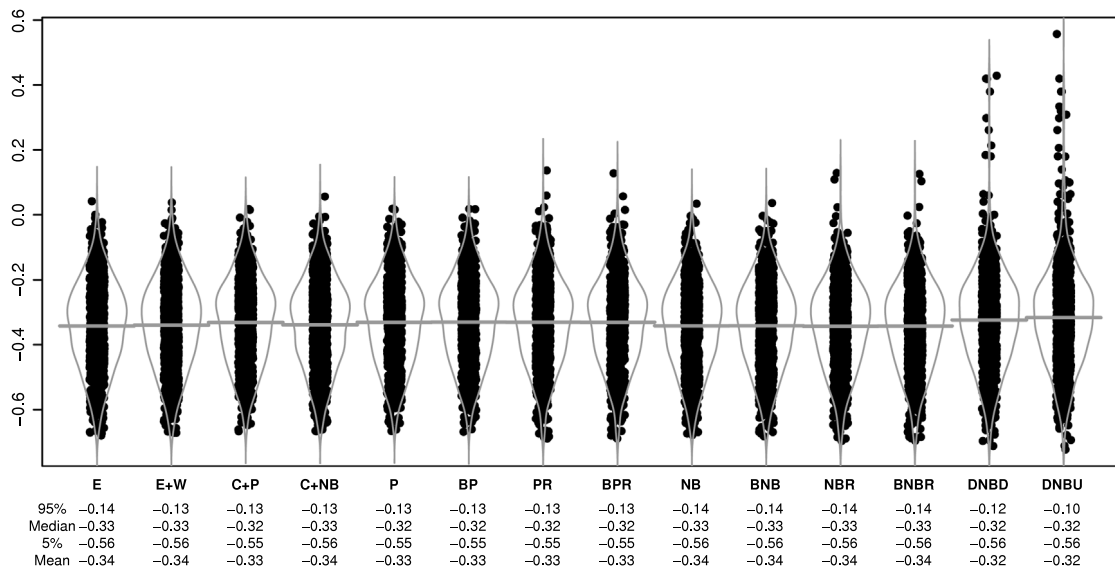


Fig. 10. Brier scores for each SKU for dataset A.

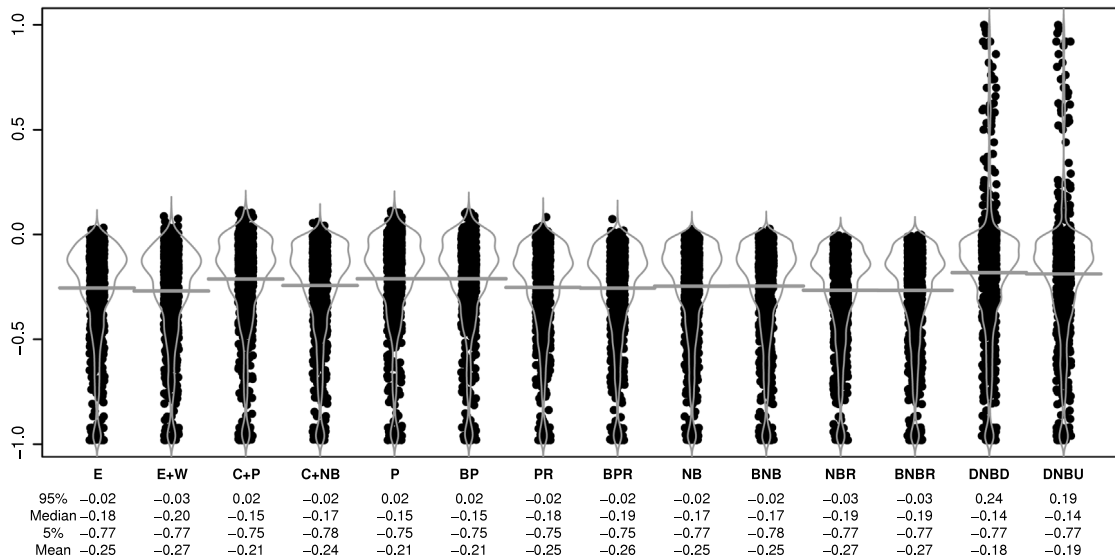


Fig. 11. Brier scores for each SKU for dataset B.

Appendix A. In addition to the results reported here, we also analyzed Fridays/Saturdays, Saturdays, days with price reductions in the evaluation period, and short-term (up to seven days ahead) forecasts. The results are reported in the [Appendix A](#).

4.7. Discussion

We note that the models in cluster 1, as per [Table 3](#), yield $\tilde{p}_t$ -value histograms that appear to be very close to uniform ([Figs. 6 and 8](#)). Nevertheless, their histograms still show a pronounced peak for $\tilde{p}_t > 0.99$, indicating that the predictive distributions still underestimate the incidence of large future values.

The models considered use different information sets that are nested to some extent: models E, P, BP, NB,

and BNB use only the raw observations, models C+P, C+NB, DNBD and DNBU also include the time ordering and model dynamics, and model E+W uses weekday information. Finally, models PR, BPR, NBR and BNBR use the full information set, including trend dynamics, weekdays, a Christmas predictor, and price information in dataset B. Intuitively, one would expect models that use larger information sets to perform better. Indeed, [Holzmann and Eulert \(2014\)](#) show that, when using correctly specified models to predict particular functionals of the predictive distribution, increasing the information set leads to lower expected values of strictly consistent scoring functions. In our case of evaluating complete predictive distributions, they show that increasing the information set will yield lower expected values of any strictly proper scoring rule.

Table 2

p-values of pairwise paired *t* tests between Brier scores for both datasets, corrected for multiple testing.

		E	E+W	C+P	C+NB	P	BP	PR	BPR	NB	BNB	NBR	BNBR	DNBD
E+W	A	*												
	B	*												
C+P	A	*	*											
	B	*	*											
C+NB	A	*	0.23	*										
	B	*		*										
P	A	*	*	0.03	*									
	B	*	*		*									
BP	A	*	*	*	*	*								
	B	*	*	*	*	1								
PR	A	*	*	1	*	1	1							
	B	0.98	*		0.0007	*	*							
BPR	A	*	*	1	*	1	1	0.07						
	B	1	*	*	*	*	*	*						
NB	A	1	0.002	*	*	*	*	*	*					
	B	*	*	*	*	*	*	0.03	*					
BNB	A	0.02	0.02	*	*	*	*	*	*	0.05				
	B	*	*	*	*	*	*	0.01	*	*				
NBR	A	1	0.001	*	*	*	*	*	*	1	0.24			
	B	*	0.09	*	*	*	*	*	*	*	*			
BNBR	A	1	0.02	*	0.002	*	*	*	*	1	1	*		
	B	*	0.10	*	*	*	*	*	*	*	*	0.98		
DNBD	A	*	*	0.02	*	0.05	0.10	0.07	0.05	*	*	*	*	
	B	*	*	0.0003	*	0.0005	0.0005	*	*	*	*	*	*	
DNBU	A	*	*		*			*	*	*	*	*	*	0.002
	B	*	*	0.008	*	0.01	0.01	*	*	*	*	*	*	0.003

* Denotes $p < 0.0001$.

Table 3

Left: clustering of models based on significantly different mean logged WT values, as per Table 1, in increasing order of mean WT. Right: clustering of models based on significantly different mean Brier scores, as per Table 2, in increasing order of mean Brier score. “C” in the table headers stands for “Cluster”, and “M” for “Model”.

WT statistics Dataset A		WT statistics Dataset B		Brier scores Dataset A		Brier scores Dataset B	
C	M	C	M	C	M	C	M
1	BNB	1	E	1	NBR	1	E+W
	E		E+W		BNBR		BNBR
	NB		BNBR	2	E		NBR
	C+NB	2	NBR		NB	2	BPR
	E+W		BNB	3	BNB		E
	BNBR		NB		E+W		PR
	NBR		C+NB	4	C+NB		NB
2	C+P	3	BPR		C+P	4	BNB
	BP		PR	5	P		C+NB
	P	5	C+P		BPR	5	C+P
	BPR		BP	6	PR		P
	PR	6	P		BP		BP
	DNBD		DNBD	7	DNBD	8	DNBU
	DNBU		DNBU		DNBU		DNBD

This is in accordance with the ordering we found for Brier scores, and in particular for dataset B (Table 3, right).

Among the models that use the smallest information sets, the very simple nonparametric Empirical approach performs surprisingly well. For dataset A, it even outperforms models using larger information sets according to Brier scores. Intra-weekly seasonality may well be visible for fast-moving SKUs and at aggregate levels, but not for the “long tail” of slow-moving products. For these products, while shoppers may indeed have a somewhat higher propensity to purchase a given product on a Friday or Saturday, this propensity may be so small that modeling it yields a worse predictive distribution than a very parsimonious model, due to the bias–variance tradeoff (Kolassa, 2016). However, we note that restricting the analysis to Saturdays (cf. Appendix A) in dataset B clearly shows the weakness of the simple Empirical approach and strongly favors approaches that do include weekday effects, namely the (Bootstrapped) NegBin Regression and Empirical + Weekdays approaches.

Similarly, modeling price reductions explicitly through negative binomial or Poisson regressions does not yield better predictive distributions than the simple Empirical model, even when the analysis is restricted to days with price reductions (cf. the Appendix A). This may be due to a misspecified price response model, or possibly

again to a high variance in the price model coefficient estimates, due to the relative rarity of price promotions, and should be investigated further. Bayesian or elastic net-type regularization may reduce the model variance without introducing too much bias, and may be an interesting avenue for further research.

We consistently find that modeling the variability using the Poisson distribution yields histogram peaks at both low and high $\tilde{p}_t$ values, indicating that the Poisson distribution underestimates the incidence of both low and high sales observations in the forecast period: true future distributions are overdispersed relative to the Poisson distribution. Overdispersion in retail sales can result from shoppers independently buying more than one unit of a given product (Ehrenberg, 1959), while zero inflation could well result from items being out-of-stock. Based on these results, we recommend that future work on retail sales forecasting should certainly not devote much effort to the modeling of sales using a Poisson model. However, we note that the Dynamic Negative Binomial models, both damped and undamped, do not perform well, especially for high quantiles. On the one hand, they do not include relevant causal factors like the day of week or the price, which may be less relevant in other datasets of intermittent demands. On the other hand, they may simply yield models that are too variable. A stronger regularization than that used here, e.g., by constraining their smoothing parameters to be much smaller (e.g., requiring that $\alpha < 0.2$ and $\alpha + \phi < 0.2$), may yield better results.

It is interesting to note that bootstrap approaches do not yield appreciably better predictive distributions than non-bootstrapped approaches. Apparently, once we have enough data, the sampling variability in parameter estimates does not make a large difference in predictive distributions, or cannot be captured adequately through bootstrapping. This finding is in line with the results of Syntetos et al. (2015), who found that the bootstrap approach of Willemain, Smart, and Schwarz (2004, 2005) did not improve stock control relative to simple parametric models. Bootstrapping may not be worth the additional runtime, which is a very relevant piece of information in the light of the large number of time series requiring forecasting in retail applications. The questions of whether bootstrapping can improve predictive distributions, and in which specific situations, merit further research.

Overall, the Brier scores paint much the same picture as the WT statistics (Table 3), though with a somewhat stronger preference for the (Bootstrapped) NegBin Regression. Figs. 10 and 11 show that the Brier scores overlap more between models than the WT statistics, but pairwise t tests extract a finer clustering.

5. Possibilities for further work

We have already indicated a few possibilities for future research, motivated by our results on our particular datasets, in the previous subsection. However, there also remain many open questions that are motivated by the rPIT and the assessment of discrete predictive distributions itself.

There exists a rich body of literature on the testing and comparison of predictive distributions for non-discrete data, mostly in applications to financial forecasting, such as stock returns, foreign exchange rates or value at risk, where the variables being forecast can be treated as approximately continuous. Examples include Amisano and Giacomini (2007), Christoffersen (1998), Diebold, Gunther, and Tay (1998), Gneiting et al. (2007), González-Rivera and Sun (2015), Mitchell and Wallis (2011); and Wallis (2003). Many approaches rely on the PIT or similar ideas of mapping the forecast errors to more manageable distributions, the null distributions of which are known and can be tested. There is definitely room here for investigating which of these techniques can be transferred to the discrete case in ways similar to that examined here. For instance, Christoffersen (1998) recommends that forecast densities be binned into equiprobable intervals, the number of realizations that fall into each bin be counted, and a standard χ^2 test applied; this approach could be applied readily to the $\tilde{p}_t$ values resulting from the rPIT.

In addition, we may also be interested in testing forecast errors for serial independence. The simplest approach, suitable for a single forecasted time series, would be to assess ACF plots of p -values (Gneiting & Katzfuss, 2014). Christoffersen (1998) tested errors in continuous predictive densities against a first order Markov chain alternative. This approach could be used on $\tilde{p}$ values that result from the rPIT described above, by discretizing them into a number of bins. As Clements and Taylor (2003) and Wallis (2003) note, Markov chains allow particular periodicities to be examined by restricting our interest to transitions between states at times t and $t + \ell$, and ignoring intervening states. For instance, in the present case of daily demands, choosing $\ell = 7$ would allow us to test for unmodeled weekly seasonality. Corradi and Swanson (2006) investigate a block bootstrap for Kolmogorov-type tests which can detect both parameter estimation error and dynamic misspecification. Alternatively, Munk, Stockis, Valeinis, and Giese (2011) propose an extension of the data-driven smooth test that was proposed by Ledwina (1994) and used here to dependent data, which could be applied directly to the output of the rPIT. Christoffersen (1998) and Wallis (2003) propose the application of a joint test of unconditional coverage and independence, which could again be adapted to count data using the rPIT. It would be interesting to investigate the suitability of proper scoring rules for detecting serial dependence in discrete predictive distribution misspecification.

One advantage of the (r)PIT over scoring rules is that it graphically shows the quantiles at which a density forecast method misspecifies the correct distribution, whereas the scoring rules examined above do not offer much guidance in this regard. An alternative representation of the ranked probability score as a quantile decomposition (Gneiting & Ranjan, 2011) offers an intriguing approach to this: we have

$$\text{rps}(\hat{f}_k, y) = \sum_{k=0}^{\infty} (\hat{F}_k - \mathbf{1}_{y \leq k})^2 = 2 \int_0^1 \text{qs}(\alpha)^{\hat{F}} d\alpha,$$

where

$$qs(\alpha)^{\hat{F}} := (1_{y \leq \hat{F}^{-1}(\alpha)} - \alpha)(\hat{F}^{-1}(\alpha) - y),$$

suggesting the use of a plot of an estimate of the integrand $qs(\alpha)^{\hat{F}}$ against the quantile α . Quantiles α where $qs(\alpha)^{\hat{F}} < qs(\alpha)^{\hat{G}}$ for predictive densities $\hat{F}$ and $\hat{G}$ indicate regions where $\hat{F}$ is better than $\hat{G}$. Figures 1, 2, 3 and 5 of Gneiting and Ranjan (2011) illustrate this idea. In addition, nonnegative weighting functions can be applied to the integrand $qs(\alpha)^{\hat{F}}$ to emphasize certain regions of the predictive distribution in the integration.

However, this approach does not work immediately in our case of many time series on different levels, because the integrand $qs(\alpha)^{\hat{F}}$ is on the order of the original time series because of the $(\hat{F}^{-1}(\alpha) - y)$ term, and therefore it is not apparent how $qs(\alpha)^{\hat{F}}$ can be compared or summarized over many series on different levels, or even for multiple forecasts of a single nonstationary (e.g., seasonal) time series. The simple expedient of dividing $qs(\alpha)^{\hat{F}}$ by the average level of each series could be used for exploratory purposes, but this “normalized” score would suffer from instability in a series with many zeros. A similar issue of incomparability between series of different levels arises with the alternative threshold decomposition proposed by Gneiting and Ranjan (2011). Of course, one could still compare the overall rps among different density forecasting models using Diebold–Mariano-type tests, as was noted above, but the potential for the quantile decomposition rps to show the exact quantiles at which one density forecast performs better than another across many time series would be an interesting avenue for future research in its own right.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <http://dx.doi.org/10.1016/j.ijforecast.2015.12.004>.

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